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Geometric semiconductor memory structure and manufacturing method thereof

Abstract

The present disclosure relates to a semiconductor structure and a manufacturing method thereof. The semiconductor structure includes: a substrate, having a first surface; a plurality of memory cells, located on the first surface of the substrate and arranged according to a first preset pattern; and a plurality of memory contact structures, corresponding to the memory cells in a one-to-one manner, where bottom portions of the memory contact structures are in contact with top portions of the memory cells, and top portions of the memory contact structures are arranged according to a second preset pattern. The bottom portion of the memory contact structure is arranged opposite to the top portion of the memory contact structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is a continuation application of International Patent Application No. PCT/CN2022/078665, filed on Mar. 1, 2022, which claims the priority to Chinese Patent Application 202111144928.3, titled “SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF” and filed with China National Intellectual Property Administration (CNIPA) on Sep. 28, 2021. The entire contents of International Patent Application No. PCT/CN2022/078665 and Chinese Patent Application 202111144928.3 are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to the technical field of integrated circuits, and in particular to a semiconductor structure and a manufacturing method thereof.

BACKGROUND

(2) Non-volatile memories, with the characteristics of high read/write speed, low power consumption, anti-radiation, and long data retention time, are irreplaceable in the field with high reliability requirements, such as national defense and aerospace.

(3) With the development of semiconductor technology, the non-volatile memory is required to have higher capacity. In order to increase the capacity of the non-volatile memory, it is necessary to arrange memory cells densely. However, the dense arrangement of memory cells limits the shape of the bit line structure and increases the production cost and process difficulty of the non-volatile memory. How to reduce the production cost and process difficulty of the non-volatile memory is an urgent problem to be solved.

SUMMARY

(4) A semiconductor structure is provided, including:

(5) a substrate, having a first surface;

(6) a plurality of memory cells, located on the first surface of the substrate and arranged according to a first preset pattern; and

(7) a plurality of memory contact structures, corresponding to the memory cells in a one-to-one manner, where bottom portions of the memory contact structures are in contact with top portions of the memory cells, and top portions of the memory contact structures are arranged according to a second preset pattern;

(8) where the bottom portion of the memory contact structure is arranged opposite to the top portion of the memory contact structure.

(9) A manufacturing method of a semiconductor structure is provided, including:

(10) providing a substrate, where the substrate has a first surface;

(11) forming a plurality of memory cells on the first surface of the substrate, where the memory cells are arranged according to a first preset pattern; and

(12) forming memory contact structures on top portions of the memory cells respectively, where top portions of the memory contact structures are arranged according to a second preset pattern.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To describe the technical solutions in the embodiments of the present disclosure or in the conventional art more clearly, the following briefly describes the accompanying drawings required for describing the embodiments or the conventional art. Apparently, the accompanying drawings in the following description show merely some embodiments of the present disclosure, and a person of ordinary skill in the art may still derive other accompanying drawings from these accompanying drawings without creative efforts.

(2) FIG. 1 is a schematic cross-sectional diagram of a semiconductor structure according to an embodiment;

(3) FIG. 2 is a schematic top view of memory cells in a semiconductor structure after the memory cells are formed according to an embodiment;

(4) FIG. 3 is a schematic top view of a semiconductor structure according to an embodiment.

(5) FIG. 4 is a schematic cross-sectional diagram of a memory cell according to an embodiment;

(6) FIG. 5 is a schematic flowchart of a manufacturing method of a semiconductor structure according to an embodiment;

(7) FIG. 6 is a schematic flowchart of a manufacturing method of a semiconductor structure according to another embodiment;

(8) FIG. 7 is a schematic flowchart of forming a transistor contact structure on each transistor according to an embodiment;

(9) FIG. 8 is a schematic cross-sectional diagram of a semiconductor structure after a transistor contact mask layer is formed according to an embodiment;

(10) FIG. 9 is a schematic cross-sectional diagram of a semiconductor structure after transistor contact structures are formed according to an embodiment corresponding to FIG. 8;

(11) FIG. 10 is a schematic cross-sectional diagram of a semiconductor structure after memory cells are formed according to an embodiment corresponding to FIG. 9;

(12) FIG. 11 is a schematic cross-sectional diagram of a semiconductor structure after a bit line mask layer is formed according to an embodiment corresponding to FIG. 10; and

(13) FIG. 12 is a schematic cross-sectional diagram of a semiconductor structure after a bit line structure is formed according to an embodiment corresponding to FIG. 11.

DETAILED DESCRIPTION

(14) To facilitate the understanding of the present disclosure, the present disclosure will be described more completely below with reference to the accompanying drawings. The embodiments of the present disclosure are shown in the drawings. However, the present disclosure may be embodied in various forms without being limited to the embodiments described herein. These embodiments are provided in order to make the present disclosure more thorough and comprehensive.

(15) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the technical field of the present disclosure. The terms mentioned herein are merely for the purpose of describing specific embodiments, rather than to limit the present disclosure.

(16) It should be understood that when an element or a layer is described as “being on”, “being adjacent to”, “being connected to” or “being coupled to” another element or layer, it can be on, adjacent to, connected to, or coupled to the another element or layer directly, or intervening elements or layers may be present. On the contrary, when an element is described as “being directly on”, “being directly adjacent to”, “being directly connected to” or “being directly coupled to” another element or layer, there are no intervening elements or layers. It should be understood that although terms such as first, second, and third may be used to describe various elements, components, regions, layers, doping types and/or sections, these elements, components, regions, layers, doping types and/or sections should not be limited by these terms. These terms are only

used to distinguish one element, component, region, layer, doped type or section from another element, component, region, layer, doped type or section. Therefore, without departing from the teachings of the present disclosure, a first element, component, region, layer, doping type or section discussed below may be a second element, component, region, layer, doping type or section. For example, the first doping type may be the second doping type, and similarly, the second doping type may be the first doping type; or the first doping type and the second doping type are different doping types, for example, the first doping type may be P-type and the second doping type may be N-type, or the first doping type may be N-type and the second doping type may be P-type.

(17) Spatial relationship terms such as “under”, “beneath”, “lower”, “below”, “above”, and “upper” can be used herein to describe the relationship shown in the figure between one element or feature and another element or feature. It should be understood that in addition to the orientations shown in the figure, the spatial relationship terms further include different orientations of used and operated devices. For example, if a device in the accompanying drawings is turned over and described as being “beneath another element”, “below it”, or “under it”, the device or feature is oriented “on” the another element or feature. Therefore, the exemplary terms “beneath” and “under” may include two orientations of above and below. In addition, the device may further include other orientations (for example, a rotation by 90 degrees or other orientations), and the spatial description used herein is interpreted accordingly.

(18) In the specification, the singular forms of “a”, “an” and “the/this” may also include plural forms, unless clearly indicated otherwise. It should also be understood that the terms such as “including/comprising” and “having” indicate the existence of the stated features, wholes, steps, operations, components, parts or combinations thereof. However, these terms do not exclude the possibility of the existence of one or more other features, wholes, steps, operations, components, parts or combinations thereof. In this case, in this specification, the term “and/or” includes any and all combinations of related listed items.

(19) The magnetic random access memory (MRAM) is a non-volatile memory. A memory cell of a memory cell in the MRAM includes a top electrode, a bottom electrode, and a magnetic material located between the top and bottom electrodes. By passing a current between the top and bottom electrodes of a target memory cell, data is written into the memory cell. The current flowing through the memory cell generates a magnetic field that can cause erroneous writes into non-target memory cells with a specific probability. The probability of an erroneous write depends on the coercive force of the memory cell (tolerance to external magnetic fields) and the magnitude of the magnetic field. The magnitude of the magnetic field is proportional to the reciprocal of the distance between the target memory cell and the affected non-target memory cell.

(20) In order to improve the integration of memory cells in the MRAM while ensuring the accuracy of data writing, the arrangement of memory cells in the MRAM and their impact on the shape or position of device structures on top of the memory cells determine the production cost and process difficulty of the MRAM.

(21) FIG. 1 is a schematic cross-sectional diagram of a semiconductor structure according to an embodiment. Referring to FIG. 1, a semiconductor structure provided by this embodiment includes: a substrate **100**, a plurality of memory cells **102**, and a plurality of memory contact structures **104**. The substrate **100** has a first surface; the memory cells **102** are located on the first surface of the substrate **100** and are arranged according to a first preset pattern. That is, the memory cells **102** are arranged on the first surface of the substrate **100** according to the first preset pattern. The memory contact structures **104** are located on the memory cells **102**, and correspond in a one-to-one manner to the memory cells **102** on the substrate **100**, and bottom portions of the memory contact structures **104** are in contact with top portions of the memory cells **102**. The top portions of the memory contact structures **104** are arranged according to a second preset pattern. That is, the top portions of the memory contact structures **104** on the memory cells **102** are arranged according to the second preset pattern; the second preset pattern and the first preset pattern are different patterns. The

bottom portion of the memory contact structure **104** is arranged opposite to the top portion of the memory contact structure **104**. That is, the bottom portion of the memory contact structure **104** is parallel to the top portion of the memory contact structure **104**.

(22) The semiconductor structure includes a plurality of memory cells **102**, located on the first surface of the substrate **100** and arranged according to the first preset pattern; and a plurality of memory contact structures **104**, corresponding to the memory cells **102** in a one-to-one manner, where the bottom portions of the memory contact structures **104** are in contact with the top portions of the memory cells **102**, and the top portions of the memory contact structures **104** are arranged according to the second preset pattern. Through such a configuration, device structures arranged according to the second preset pattern can be formed on the memory cells **102** arranged according to the first preset pattern, which eliminates the impact of the arrangement shape of the memory cells **102** on the arrangement or shape of the device structures formed on the memory cells **102**, thus reducing the process difficulty and the production cost.

(23) FIG. 2 is a schematic top view of memory cells in a semiconductor structure after the memory cells **102** are formed according to an embodiment. As shown in FIG. 2, in an embodiment, the first preset pattern includes a regular hexagon, and the memory cells **102** are located at vertex positions and a central position of the regular hexagon.

(24) In an embodiment, the bottom portions of the memory contact structure **104** are arranged according to the first preset pattern.

(25) In an embodiment, an area of the bottom portion of the memory contact structure **104** is larger than that of the top portion of the memory cell **102**. In other embodiments, the area of the bottom portion of the memory contact structure **104** is smaller than or equal to the area of the top portion of the memory cell **102**. Such a configuration allows the memory contact structure **104** to be in close contact with the memory cell **102** while avoiding damage to the memory cell **102** in the process of forming the memory contact structure **104**.

(26) In an embodiment, a material of the memory contact structure **104** includes one or more selected from a group consisting of polysilicon, a metal, a conductive metal nitride, a conductive metal oxide, and a metal silicide. The metal may be tungsten (W), nickel (Ni), or titanium (Ti); the conductive metal nitride includes titanium nitride (TiN); the conductive metal oxide includes iridium oxide (IrO₂); the metal silicide includes titanium silicide (TiSi).

(27) FIG. 3 is a schematic top view of a semiconductor structure according to an embodiment. As shown in FIG. 3, in an embodiment, the second preset pattern includes an array with a plurality of rows and columns.

(28) Further referring to FIG. 1, in an embodiment, the semiconductor structure further includes: a plurality of bit line structures **106**, where any one of the bit line structures **106** is in contact with the top portion of the memory contact structure **104** in a same column, and the bit line structure **106** is a straight-line shape. For example, a length of the bit line structure **106** along a direction X is not less than a length of the top portion of the memory contact structure **104** along the direction X, where the direction X intersects with an extension direction of the bit line structure **106**.

(29) Further referring to FIG. 1, in an embodiment, the semiconductor structure further includes: a plurality of transistors **108** and a plurality of transistor contact structures **110**. The transistors **108** are located between the substrate **100** and the memory cells **102**, and correspond to the memory cells **102** in a one-to-one manner. The transistor contact structures **110** are located between the transistors **108** and the memory cells **102**, and are in contact with the transistors **108** and the memory cells **102**. The top portions of the transistor contact structures **110** are arranged according to the first preset pattern.

(30) In an embodiment, the transistors **108** are arranged according to a third preset pattern. It may be understood that, the third preset pattern includes the first preset pattern and the second preset pattern. In practical application, the arrangement shape of the transistors **108** may be set as required. For example, when the transistors **108** are arranged according to the first preset pattern,

the transistors **108** and the memory cells **102** are arranged in the same manner. When the transistors **108** are arranged according to the second preset pattern, the transistors **108** and the memory contact structures **104** are arranged in the same manner.

(31) In an embodiment, bottom portions of the transistor contact structures **110** are arranged according to the third preset pattern. The bottom portion of the transistor contact structure **110** is arranged opposite to a top portion of the transistor contact structure **110**. The third preset pattern includes the first preset pattern and the second preset pattern.

(32) In an embodiment, an area of the top portion of the transistor contact structure **110** is larger than that of the bottom portion of the memory cell **102**. In other embodiments, the area of the top portion of the transistor contact structure **110** is smaller than or equal to that of the bottom portion of the memory cell **102**, so as to eliminate impact of the process deviation on the contact resistance between the transistor contact structure **110** and the memory cell **102**.

(33) Further referring to FIG. **1**, in an embodiment, the transistor contact structures **110** each include: a first lead-out structure **202** and a second lead-out structure **204**. A bottom portion of the first lead-out structure **202** is in contact with the transistor **108** to lead out the transistor **108**. For example, the bottom portion of the first lead-out structure **202** is connected to a drain of the transistor **108**, to lead out the drain of the transistor **108**. Typically, an area of the bottom portion of the first lead-out structure **202** is equal to that of the drain of the transistor **108**. A bottom portion of the second lead-out structure **204** is in contact with a top portion of the first lead-out structure **202**. The bottom portion of the first lead-out structure **202** is arranged opposite to the top portion of the first lead-out structure **202**, and the bottom portion of the second lead-out structure **204** is arranged opposite to the top portion of the second lead-out structure **204**. The bottom portion of the first lead-out structure **202** is the bottom portion of the transistor contact structure **110**, and the top portion of the second lead-out structure **204** is the top portion of the transistor contact structure **110**. In this case, the bottom portions of the first lead-out structures **202** are arranged according to the third preset pattern, and the top portions of the second lead-out structures **204** are arranged according to the first preset pattern. For example, the bottom portions of the first lead-out structure **202** are arranged according to the second preset pattern, and the top portions of the second lead-out structure **204** are arranged according to the first preset pattern; alternatively, the bottom portions of the first lead-out structure **202** and the top portions of the second lead-out structures **204** are both arranged according to the first preset pattern.

(34) In an embodiment, an area of the bottom portion of the second lead-out structure **204** is not smaller than that of the top portion of the first lead-out structure **202**.

(35) In an embodiment, a portion of the first lead-out structure **202** which is close to the bottom portion of the second lead-out structure **204** is a trapezoidal structure.

(36) In an embodiment, a material of the first lead-out structure **202** and the second lead-out structure **204** includes one or more selected from a group consisting of polysilicon, a metal, a conductive metal nitride, a conductive metal oxide, and a metal silicide. The metal may be tungsten (W), nickel (Ni), or titanium (Ti); the conductive metal nitride includes titanium nitride (TiN); the conductive metal oxide includes iridium oxide (IrO₂); the metal silicide includes titanium silicide (TiSi). For example, the material of at least one of the memory contact structure **104**, the first lead-out structure **202**, and the second lead-out structure **204** is different from others. For example, the material of the first lead-out structure **202** is different from the material of the memory contact structure **104** and the second lead-out structure **204**. In practical application, a same material may be selected as required to manufacture the memory contact structure **104**, the first lead-out structure **202**, and the second lead-out structure **204**.

(37) In an embodiment, the memory cell **102** includes a ferroelectric memory cell, a magnetoresistive memory cell, a resistance change memory cell, or a phase change memory cell.

(38) FIG. **4** is a schematic cross-sectional diagram of a memory cell **102** according to an embodiment. As shown in FIG. **4**, in this embodiment, the memory cell **102** is a magnetoresistive

memory cell. The memory cell **102** includes: a fixed layer **206**, a non-magnetic isolation layer **208**, and a free layer **210**. The fixed layer **206** is located at the bottom portion of the memory cell **102**. That is, the fixed layer **206** is connected to the transistor contact structure **110** and has a magnetic field of a preset direction. Compared with the free layer **210**, the fixed layer **206** has a relatively large film thickness and relatively strong magnetic properties, and the magnetic torque is not easily reversed. For example, the material of the fixed layer **206** includes CoFe and CoFeB. Typically, the fixed layer **206** is a laminated structure consisting of multiple films. The laminated structure includes a seed layer, a [Co(x)/Pt(y)].sub.m laminated layer, a Ru or Ir metal layer, a [Co(x)/Pt(y)].sub.m laminated layer, a Ta metal layer, and a CoFeB metal layer in sequence from bottom to top. The non-magnetic isolation layer **208** is located between the free layer **210** and the fixed layer **206**, to isolate the fixed layer **206** from the free layer **210**. For example, the material of the non-magnetic isolation layer **208** includes MgO and Al.sub.2O.sub.3. The free layer **210** is made of a soft ferromagnetic material, which has low coercivity, high magnetic permeability, and high sensitivity to low magnetic fields. The free layer **210** has weak magnetic properties and the magnetic torque is easily reversed. For example, the material of the free layer **210** includes CoFe, NiFe, NiFeCo, and CoFeB.

(39) The magnetoresistive memory cell is a current-controlled element, where a magnetization direction of the free layer **210** is controlled by a current flowing through the memory cell **102**. When the magnetization direction of the free layer **210** is the same as that of the fixed layer **206**, the memory cell **102** is in a low-resistance state, and the memory cell **102** stores data “0”. When the magnetization direction of the free layer **210** is opposite to that of the fixed layer **206**, the memory cell **102** is in a high-resistance state, and the memory cell **102** stores data “1”.

(40) Further referring to FIG. 4, in an embodiment, the memory cell **102** further includes a bottom electrode **212**, which is located on an upper surface of the transistor contact structure **110**. It may be understood that, in some embodiments, the second lead-out structure **204** may also be used as the bottom electrode **212** of the memory cell **102**.

(41) Further referring to FIG. 4, in an embodiment, the memory cell **102** further includes a top electrode **214**, which is located on an upper surface of the free layer **210**. It may be understood that, in some embodiments, the memory contact structure **104** may also be used as the top electrode **214** of the memory cell **102**.

(42) Further referring to FIG. 4, in other embodiments, the memory cell **102** further includes a protective layer **216**, covering a sidewall of the fixed layer **206**, and the sidewall of the fixed layer **206** extends to cover the sidewall of the free layer **210**. For example, a material of the protective layer **216** is silicon nitride.

(43) Further referring to FIG. 1, in an embodiment, the semiconductor structure further includes: a plurality of word line structures **112** and a plurality of global source lines **114**, where the word line structure **112** is connected to a gate of the transistor **108**, and the global source line **114** is connected to a source of the transistor **108**.

(44) FIG. 5 is a schematic flowchart of a manufacturing method of a semiconductor structure according to an embodiment. As shown in FIG. 1 and FIG. 5, the present disclosure further provides a manufacturing method of a semiconductor structure, including the following steps.

(45) **S102**: Provide a substrate, where the substrate has a first surface.

(46) **S104**: Form a plurality of memory cells arranged according to a first preset pattern on the first surface of the substrate.

(47) As shown in FIG. 1, a plurality of memory cells **102** are formed on the first surface of the substrate **100**, and the memory cells **102** are arranged on the first surface of the substrate **100** according to the first preset pattern.

(48) **S106**: Form memory contact structures on top portions of the memory cells respectively, where top portions of the memory contact structures are arranged according to a second preset pattern.

(49) Specifically, the memory contact structures **104** are formed on the memory cells **102** respectively. Bottom portions of the memory contact structures **104** are in contact with the top portions of the memory cell **102**. The top portions of the memory contact structures **104** are arranged according to the second preset pattern. The second preset pattern and the first preset pattern are different patterns. The bottom portion of the memory contact structure **104** is arranged opposite to the top portion of the memory contact structure **104**. That is, the bottom portion of the memory contact structure **104** is parallel to the top portion of the memory contact structure **104**.

(50) The foregoing manufacturing method of a semiconductor structure includes providing a substrate **100** having a first surface, forming a plurality of memory cells **102** arranged according to a first preset pattern on the first surface of the substrate **100**, and forming memory contact structures **104** on top portions of the memory cells **102** respectively, where top portions of the memory contact structures **104** are arranged according to a second preset pattern. Through such a configuration, device structures arranged according to the second preset pattern can be formed on the memory cells **102** arranged according to the first preset pattern, which eliminates the impact of the arrangement shape of the memory cells **102** on the arrangement or shape of the device structures formed on the memory cells **102**, thus reducing the process difficulty and the production cost.

(51) As shown in FIG. 2, in an embodiment, the first preset pattern includes a regular hexagon, and the memory cells **102** are located at vertex positions and a central position of the regular hexagon.

(52) In an embodiment, the bottom portion of the memory contact structure **104** is in contact with the top portion of the memory cell **102**, and the bottom portions of the memory contact structures **104** are arranged according to the first preset pattern.

(53) In an embodiment, an area of the bottom portion of the memory contact structure **104** is larger than an area of the top portion of the memory cell **102**. In other embodiments, the area of the bottom portion of the memory contact structure **104** is smaller than or equal to the area of the top portion of the memory cell **102**. Such a configuration allows the memory contact structure **104** to be in close contact with the memory cell **102** while avoiding damage to the memory cell **102** in the process of forming the memory contact structure **104**.

(54) In an embodiment, a material of the memory contact structure **104** includes one or more selected from a group consisting of polysilicon, a metal, a conductive metal nitride, a conductive metal oxide, and a metal silicide. The metal may be tungsten (W), nickel (Ni), or titanium (Ti); the conductive metal nitride includes titanium nitride (TiN); the conductive metal oxide includes iridium oxide (IrO₂); the metal silicide includes titanium silicide (TiSi).

(55) As shown in FIG. 3, in an embodiment, the second preset pattern includes an array with a plurality of rows and columns.

(56) Further referring to FIG. 1, in an embodiment, the manufacturing method of a semiconductor structure further includes:

(57) forming a plurality of bit line structures **106** on the top portions of the memory contact structures **104** respectively, where any one of the bit line structures **106** is in contact with the top portion of the memory contact structure **104** in a same column, and the bit line structure **106** is a straight-line shape. For example, a length of the bit line structure **106** along a direction X is not less than a length of the top portion of the memory contact structure **104** along the direction X, where the direction X intersects with an extension direction of the bit line structure **106**.

(58) FIG. 6 is a schematic flowchart of a manufacturing method of a semiconductor structure according to another embodiment. As shown in FIG. 1 and FIG. 6, in an embodiment, before step S104, the method further includes the following steps:

(59) S202: Form a plurality of transistors on the first surface of the substrate.

(60) Specifically, a plurality of transistors **108** are formed on the first surface of the substrate **100**, where the transistors **108** correspond to the memory cells **102** in a one-to-one manner. In an embodiment, the transistors **108** are arranged according to a third preset pattern. It may be

understood that, the third preset pattern includes the first preset pattern and the second preset pattern. In practical application, the arrangement shape of the transistors **108** may be set as required. For example, when the transistors **108** are arranged according to the first preset pattern, the transistors **108** and the memory cells **102** are arranged in the same manner. When the transistors **108** are arranged according to the second preset pattern, the transistors **108** and the memory contact structures **104** are arranged in the same manner.

(61) **S204**: Form transistor contact structures on the transistors respectively, where top portions of the transistor contact structures are arranged according to the first preset pattern.

(62) Specifically, the transistor contact structures **110** are formed on the transistors **108** respectively. The transistor contact structures **110** are in contact with the transistors **108** and the memory cells **102**, and the top portions of the transistor contact structures **110** are arranged according to the first preset pattern.

(63) In an embodiment, bottom portions of the transistor contact structures **110** are arranged according to the third preset pattern. The bottom portion of the transistor contact structure **110** is arranged opposite to a top portion of the transistor contact structure **110**. The third preset pattern includes the first preset pattern and the second preset pattern.

(64) In an embodiment, an area of the top portion of the transistor contact structure **110** is larger than that of the bottom portion of the memory cell **102**. In other embodiments, the area of the top portion of the transistor contact structure **110** is smaller than or equal to that of the bottom portion of the memory cell **102**, so as to eliminate impact of the process deviation on the contact resistance between the transistor contact structure **110** and the memory cell **102**.

(65) FIG. 7 is a schematic flowchart of forming a transistor contact structure on each transistor according to an embodiment. As shown in FIG. 1 and FIG. 7, in an embodiment, the transistor contact structures **110** each include a first lead-out structure **202** and a second lead-out structure **204**. Step **S204** includes the following steps:

(66) **S302**: Form the first lead-out structures **202** on the transistors **108** respectively, where bottom portions of the first lead-out structures **202** are in contact with the transistors **108**.

(67) Specifically, the bottom portion of the first lead-out structure **202** is in contact with the transistor **108** to lead out the transistor **108**. For example, the bottom portion of the first lead-out structure **202** is connected to a drain of the transistor **108** to lead out the drain of the transistor **108**. Typically, an area of the bottom portion of the first lead-out structure **202** is equal to that of the drain of the transistor **108**.

(68) **S304**: Form the second lead-out structures on top portions of the first lead-out structures respectively, where top portions of the second lead-out structures are in contact with the bottom portions of the memory cells.

(69) Specifically, the bottom portion of the first lead-out structure **202** is the bottom portion of the transistor contact structure **110**, and the top portion of the second lead-out structure **204** is the top portion of the transistor contact structure **110**. In this case, the bottom portions of the first lead-out structures **202** are arranged according to the third preset pattern, and the top portions of the second lead-out structures **204** are arranged according to the first preset pattern. For example, the bottom portions of the first lead-out structure **202** are arranged according to the second preset pattern, and the top portions of the second lead-out structure **204** are arranged according to the first preset pattern; alternatively, the bottom portions of the first lead-out structure **202** and the top portions of the second lead-out structures **204** are both arranged according to the first preset pattern.

(70) FIG. 8 is a schematic cross-sectional diagram of a semiconductor structure after a transistor contact mask layer is formed according to an embodiment; FIG. 9 is a schematic cross-sectional diagram of a semiconductor structure after transistor contact structures are formed according to an embodiment corresponding to FIG. 8.

(71) As shown in FIG. 8 and FIG. 9, in the first step, first contact structures **302**, a first conductive film **304**, and a transistor contact mask layer **306** are sequentially formed on the substrate **100** on

which the transistors **108** have been formed. The first contact structure **302** is connected to the drain of the transistor **108**, and a first dielectric layer **308** is filled between adjacent first contact structures **302**, where an upper surface of the first dielectric layer **308** is flush with an upper surface of the first contact structure **302**. The first conductive film **304** is located on the upper surface of the first contact structure **302**, and the transistor contact mask layer **306** is located on the first conductive film **304**, to define the shape and position of the transistor contact structure **110**. In the second step, a part of the first conductive film **304**, a part of the first contact structure **302**, and a part of the first dielectric layer **308** are first removed through an etching process by using the transistor contact mask layer **306** as a mask, to obtain the first lead-out structure **202** formed by the remaining first contact structure **302**, and the second lead-out structure **204** formed by the remaining first conductive film **304**. It may be understood that, in some embodiments, only the first conductive film **304** and the first contact structure **302** that are not covered by the transistor contact mask layer **306** are removed through an etching process, to form the second lead-out structure **204** and the first lead-out structure **202**. Next, a second dielectric layer **310** is filled between adjacent second lead-out structures **204**, where an upper surface of the second dielectric layer **310** is flush with an upper surface of the second lead-out structure **204**.

(72) In an embodiment, an area of a bottom portion of the second lead-out structure **204** is not smaller than that of a top portion of the first lead-out structure **202**, where the bottom portion of the second lead-out structure **204** is arranged opposite to the top portion of the second lead-out structure **204**.

(73) In an embodiment, a portion of the first lead-out structure **202** which is close to the bottom portion of the second lead-out structure **204** is a trapezoidal structure.

(74) In an embodiment, a material of the first lead-out structure **202** and the second lead-out structure **204** includes one or more selected from a group consisting of polysilicon, a metal, a conductive metal nitride, a conductive metal oxide, and a metal silicide. The metal may be tungsten (W), nickel (Ni), or titanium (Ti); the conductive metal nitride includes titanium nitride (TiN); the conductive metal oxide includes iridium oxide (IrO₂); the metal silicide includes titanium silicide (TiSi). For example, the material of at least one of the memory contact structure **104**, the first lead-out structure **202**, and the second lead-out structure **204** is different from others. For example, the material of the first lead-out structure **202** is different from the material of the memory contact structure **104** and the second lead-out structure **204**. In practical application, a same material may be selected as required to manufacture the memory contact structure **104**, the first lead-out structure **202**, and the second lead-out structure **204**.

(75) In an embodiment, the memory cell **102** includes a ferroelectric memory cell, a magnetoresistive memory cell, a resistance change memory cell, or a phase change memory cell.

(76) In an embodiment, the bottom portions of the first lead-out structure **202** are arranged according to the second preset pattern, and the top portions of the second lead-out structure **204** are arranged according to the first preset pattern.

(77) FIG. **10** is a schematic cross-sectional diagram of a semiconductor structure after memory cells are formed according to an embodiment corresponding to FIG. **9**; FIG. **11** is a schematic cross-sectional diagram of a semiconductor structure after a bit line mask layer is formed according to an embodiment corresponding to FIG. **10**; and FIG. **12** is a schematic cross-sectional diagram of a semiconductor structure after a bit line structure is formed according to an embodiment corresponding to FIG. **11**. As shown in FIG. **10**, FIG. **11**, and FIG. **12**, in the third step, memory cells **102** are formed on the second lead-out structures **204** respectively. For example, if the memory cell **102** is a magnetoresistive memory cell, the memory cell **102** is formed through the following steps: First, a fixed layer structure is formed on the upper surface of the second lead-out structure **204** through a common deposition process, such as a physical vapor deposition process. For example, the fixed layer structure is a laminated structure consisting of a plurality of films, including a seed layer, a [Co(x)/Pt(y)]_{sub.m} laminated layer, a Ru or Ir metal layer, a

[Co(x)/Pt(y)].sub.m laminated layer, a Ta metal layer, and a CoFeB metal layer that are sequentially formed on the upper surface of the second lead-out structure **204**. Next, a non-magnetic isolation structure and a free layer structure are sequentially formed on the upper surface of the fixed layer structure. For example, the material of the non-magnetic isolation structure includes MgO and Al.sub.2O.sub.3; the material of the free layer structure includes CoFe, NiFe, NiFeCo, and CoFeB. Then, an annealing process is performed, and excess parts of the free layer structure, the non-magnetic isolation structure, and the fixed layer structure are removed through a photoetching process, to obtain the free layer **210** formed by the remaining free layer structure, the non-magnetic isolation layer **208** formed by the remaining non-magnetic isolation structure, and the fixed layer **206** formed by the remaining fixed layer structure. Typically, an area of the bottom portion of the fixed layer **206** is equal to that of the top portion of the second lead-out structure **204**. Then, a protection structure, such as a silicon nitride structure, is formed on sidewalls of the fixed layer **206**, the non-magnetic isolation layer **208**, and the free layer **210**. The protection structure extends along the sidewall of the free layer **210** to cover the upper surface of the free layer **210**. Next, third dielectric structures are formed on the substrate **100**, where each third dielectric structure is filled between adjacent free layers **210**, and an upper surface of the third dielectric structure is higher than an upper surface of the protection structure. Further, the protection structure and the third dielectric structure that are right above the free layer **210** are removed through a photoetching process, to obtain a memory contact trench **312** above the free layer **210**, a protective layer **216** formed by the remaining protection structure, and a third dielectric layer **314** formed by the remaining third dielectric structure. In the fourth step, a memory contact layer **316** is filled in the memory contact trench **312**, where an upper surface of the memory contact layer **316** is flush with an upper surface of the third dielectric layer **314**. In the fifth step, a bit line structure layer **318** is formed on the upper surface of the third dielectric layer **314**. In the sixth step, a bit line mask layer **320** is formed on the bit line structure layer, where the bit line mask layer **320** defines shapes and positions of the bit line structure **106** and the memory contact structure **104**. In other embodiments, before the fifth step, the method includes: before forming, on the upper surface of the third dielectric layer **314**, a memory mask layer for defining the shape and position of the memory contact structure **104**, removing a part of the memory contact layer **316** through a photoetching process, to obtain the memory contact structure **104** formed by the remaining memory contact layer **316**. In this case, the bit line mask layer **320** is only used for defining the shape and position of the bit line structure **106**. In the seventh step, a part of the bit line structure layer, a part of the memory contact layer **316**, and a part of the third dielectric layer **314** are removed through an etching process by using the bit line mask layer **320** as a mask, to obtain the bit line structure **106** formed by the remaining bit line structure layer and the memory contact structure **104** formed by the remaining memory contact layer **316**. In the eighth step, a fourth dielectric layer **322** is filled between adjacent bit line structures **106**, where an upper surface of the fourth dielectric layer **322** is flush with the bit line structure. For example, materials of the first dielectric layer **308**, the second dielectric layer **310**, the third dielectric layer **314**, and the fourth dielectric layer **322** include silicon dioxide, silicon oxynitride, and silicon nitride.

(78) Embodiments of the present disclosure are described herein with reference to cross-sectional views as schematic diagrams of idealized embodiments (and intermediate structures) of the present disclosure, such that variations shown in the shapes and due to, for example, manufacturing techniques and/or tolerances can be contemplated. Therefore, the embodiments of the present disclosure should not be limited to the specific shapes of the regions shown herein, but include shape deviations due to, for example, manufacturing techniques. For example, an injection region displayed as a rectangle usually has a circular or curved feature and/or injection concentration gradient at an edge of the region, rather than a binary change from the injection region to a non-injection region. Similarly, a buried region formed by injection can lead to some injection in a region between the buried region and a surface through which the injection is carried out.

Therefore, the regions shown in the figure are schematic in nature, and their shapes are not intended to show the actual shapes of the regions of the device and limit the scope of the present disclosure.

(79) In the specification, the description of terms such as “some embodiments”, “other embodiments”, “desirable embodiments” and the like means that a specific feature, structure, material or characteristic described in combination with the embodiment(s) or example(s) are included in at least one embodiment or example of the present application. In this specification, the schematic description of the above terms does not necessarily refer to the same embodiment or example.

(80) The technical characteristics of the foregoing embodiments can be employed in arbitrary combinations. To provide a concise description of these embodiments, not all possible combinations of all technical characteristics of the embodiments are described; however, these combinations of technical characteristics should be construed as disclosed in the description as long as no contradiction occurs.

(81) Only several embodiments of the present application are described in detail above, but they should not therefore be construed as limiting the scope of the present application. It should be noted that those of ordinary skill in the art can further make variations and improvements without departing from the conception of the present application. These variations and improvements all fall within the protection scope of the present application. Therefore, the protection scope of the present disclosure should be subject to the protection scope defined by the claims.

Claims

1. A semiconductor structure, comprising: a substrate, having a first surface; a plurality of memory cells, located above the first surface of the substrate and arranged according to a first preset pattern; a plurality of transistors, located between the substrate and the memory cells, the transistors correspond to the memory cells in a one-to-one manner, and the transistors are arranged according to a second preset pattern different to the first preset pattern, the second preset pattern comprises an array with a plurality of rows and columns; a plurality of transistor contact structures, located between the transistors and the memory cells and in contact with the transistors and the memory cells, wherein top portions of the transistor contact structures are arranged according to the first preset pattern, bottom portions of the transistor contact structures are arranged according to the second preset pattern; and a plurality of memory contact structures, corresponding to the memory cells in a one-to-one manner, wherein bottom portions of the memory contact structures are in contact with top portions of the memory cells, and top portions of the memory contact structures are arranged according to the second preset pattern; wherein the bottom portions of the memory contact structures are arranged opposite to the top portions of the memory contact structures.
2. The semiconductor structure according to claim 1, wherein the first preset pattern comprises a regular hexagon, and the memory cells are located at vertex positions and a central position of the regular hexagon.
3. The semiconductor structure according to claim 1, wherein the bottom portions of the memory contact structures are arranged according to the first preset pattern.
4. The semiconductor structure according to claim 1, the semiconductor structure further comprises: a plurality of bit line structures, wherein any one of the bit line structures is in contact with the top portion of the memory contact structure in a same column, and the bit line structure is in a straight-line shape.
5. The semiconductor structure according to claim 1, wherein the bottom portion of the transistor contact structure is arranged opposite to the top portion of the transistor contact structure.
6. The semiconductor structure according to claim 1, wherein the transistor contact structures each comprise: a first lead-out structure, wherein a bottom portion of the first lead-out structure is in

contact with the transistor; and a second lead-out structure, wherein a bottom portion of the second lead-out structure is in contact with a top portion of the first lead-out structure, and a top portion of the second lead-out structure is the top portion of the transistor contact structure; wherein the bottom portion of the first lead-out structure is arranged opposite to the top portion of the first lead-out structure, and the bottom portion of the second lead-out structure is arranged opposite to the top portion of the second lead-out structure.

7. The semiconductor structure according to claim 6, wherein an area of the bottom portion of the second lead-out structure is not smaller than that of the top portion of the first lead-out structure.

8. The semiconductor structure according to claim 1, wherein the memory cell comprises a ferroelectric memory cell, a magnetoresistive memory cell, a resistance change memory cell, or a phase change memory cell.

9. A manufacturing method of a semiconductor structure, comprising: providing a substrate, wherein the substrate has a first surface; forming a plurality of transistors on the first surface of the substrate, wherein the transistors are arranged according to a second preset pattern different to the first preset pattern, the second preset pattern comprises an array with a plurality of rows and columns; forming transistor contact structures on the transistors respectively, wherein top portions of the transistor contact structures are arranged according to the first preset pattern, bottom portions of the transistor contact structures are arranged according to the second preset pattern; forming a plurality of memory cells above the first surface of the substrate, wherein the memory cells are arranged according to a first preset pattern, wherein the transistor contact structures are in contact with the transistors and the memory cells, and the transistors correspond to the memory cells in a one-to-one manner; and forming memory contact structures on top portions of the memory cells respectively, wherein top portions of the memory contact structures are arranged according to a second preset pattern.

10. The manufacturing method according to claim 9, wherein the first preset pattern comprises a regular hexagon, and the memory cells are located at vertex positions and a central position of the regular hexagon.

11. The manufacturing method according to claim 9, wherein bottom portions of the memory contact structures are in contact with the top portions of the memory cells respectively, and the bottom portions of the memory contact structures are arranged according to the first preset pattern; and the bottom portion of the memory contact structure is arranged opposite to the top portion of the memory contact structure.

12. The manufacturing method according to claim 9, the manufacturing method further comprises: forming a plurality of bit line structures on the top portions of the memory contact structures respectively, wherein any one of the bit line structures is in contact with the top portion of the memory contact structure in a same column, and the bit line structure is a straight-line shape.

13. The manufacturing method according to claim 9, wherein the bottom portion of the transistor contact structure is arranged opposite to the top portion of the transistor contact structure.

14. The manufacturing method according to claim 9, wherein the transistor contact structures each comprise a first lead-out structure and a second lead-out structure, and the forming transistor contact structures on the transistors respectively comprises: forming the first lead-out structures on the transistors respectively, wherein bottom portions of the first lead-out structures are in contact with the transistors; and forming the second lead-out structures on top portions of the first lead-out structures respectively, wherein top portions of the second lead-out structures are in contact with the bottom portions of the memory cells; wherein the bottom portion of the first lead-out structure is arranged opposite to the top portion of the first lead-out structure.

15. The manufacturing method according to claim 14, wherein an area of the bottom portion of the second lead-out structure is not smaller than that of the top portion of the first lead-out structure; and the bottom portion of the second lead-out structure is arranged opposite to the top portion of the second lead-out structure.

16. The manufacturing method according to claim 15, wherein the bottom portions of the first lead-out structures are arranged according to the second preset pattern, and the top portions of the second lead-out structures are arranged according to the first preset pattern.
